

1 Quick refresher on spin

Spin can be described with the operators $\hat{S}_x, \hat{S}_y, \hat{S}_z$ that obey the following commutation relations:

$$\boxed{[\hat{S}_j, \hat{S}_k] = i\hbar \epsilon_{jkl} \hat{S}_\ell} \quad (\hbar = 1), \quad (1)$$

with $i, j, k = x, y, z = 1, 2, 3$ and using the Einstein summation convention. One can define raising and lowering operators as

$$\boxed{\hat{S}_\pm = \hat{S}_x \pm i\hat{S}_y} \quad \text{with} \quad \boxed{[S_-, S_+] = -2S_z}. \quad (2)$$

Proof.

$$\begin{aligned} [S_-, S_+] &= [S_x - iS_y, S_x + iS_y] \\ &= [S_x, S_x] + i[S_x, S_y] - i[S_y, S_x] + [S_y, S_y] \\ &= 2i[S_x, S_y] \stackrel{(1)}{=} 2i \cdot i \epsilon_{12\ell} S_\ell = -2 \epsilon_{12\ell} S_\ell = -2S_z, \end{aligned} \quad (3)$$

where the first step used the bilinearity of the commutator and the last step uses $\epsilon_{jkl} = 0$ if two indices are equal, leaving $\epsilon_{123} S_3 = S_z$. $\square$

For spin- $\frac{1}{2}$ particles, one can write

$$S_j = \frac{\hbar}{2} \sigma_j \quad \text{with } \sigma_j \text{ the Pauli matrices, } j = x, y, z, \quad (4)$$

and often again $\hbar = 1$.